EE 315 Assignment 10

Transient Stability Assessment of a Power System and Identification of Critical Clearing Time

Course	EE 315 - Power Systems
Assignment	Project / Assignment 10
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Group	04
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Project Code	`Assignment10_Akarsh_G4_M1.m`

1 Group Members

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2 Introduction

The objective of this project is to model a nine-bus power system and assess its transient rotor-angle stability following a single-line-to-ground fault. A MATLAB program determines the initial operating point, simulates generator rotor motion for different fault durations, and searches for a critical clearing-time bracket. The fault resistance is selected from the last digit of roll number 240002007, giving **Rf = 7 ohm**.

3 System Model and Fault Description

The study uses the three-generator, nine-bus MATPOWER case9 network. A phase-a-to-ground fault is applied on line 7-8, at a point 1% of the line length from bus 8. It starts at 0.100 s and is cleared by opening line 7-8 at both ends, without reclosure. Classical generators have constant mechanical input and constant internal-voltage magnitudes.

4 Software Implementation & Code Availability

Run `results = Assignment10_Akarsh_G4_M1;` in MATLAB. The standalone function saves trajectories, a duration scan and plots in `results_matlab/`. A matching static website is supplied for GitHub Pages. Its JavaScript solver executes in the browser; the MATLAB file is downloaded separately.

5 Formulas Used

5.1 Lecture Swing Equation

$$\frac{H_i}{\pi f}\frac{d^2\delta_i}{dt^2} = P_{m,\,i} - P_{e,\,i}, \qquad H_{i,\,\mathrm{sys}} = H_{i,\,\mathrm{mach}}\frac{S_{i,\,\mathrm{mach}}}{S_{\mathrm{base}}}$$

Lecture 30, pages 3-4: angles are electrical radians, powers and inertia use the common system base. The damping term is zero. Page 7 supplies the constant mechanical-input, constant internal-voltage and constant-admittance-load assumptions. Line and machine resistances are ignored. The specified fault resistance remains present.

5.2 Per-Unit Fault Resistance

$$Z_{\mathrm{base}} = \frac{345^2}{100} = 1190.25\ \Omega, \qquad R_{f,\,\mathrm{pu}} = \frac{7}{1190.25} = 0.00588112$$

5.3 Single-Line-to-Ground Fault

Lecture 27 connects the sequence networks in series: I0 = I1 = I2 and Ia = 3I1. Eliminating the passive negative- and zero-sequence networks gives the equivalent shunt at the positive-sequence fault point:

$$Y_{f,\,\mathrm{eq}} = \frac{1}{Z_{2,\,ff} + Z_{0,\,ff} + 3R_{f,\,\mathrm{pu}}}$$

The positive-sequence driving-point impedance is already in its network. It must not be added again in this shunt.

5.4 Network Calculation

$$E_i' = V_i + jX_{di}'I_i, \qquad Y_{L,\,i} = \frac{S_{L,\,i}^*}{|V_i|^2}, \qquad Y_{\mathrm{red}} = Y_{gg} - Y_{gb}Y_{bb}^{-1}Y_{bg}$$

$$P_{e,\,i} = \mathrm{Re}\left\{ E_i \left(\sum_j Y_{\mathrm{red},\,ij} E_j \right)^* \right\}$$

Nodal elimination, load-flow iteration and RK4 are numerical implementation procedures. They are not claimed to be formulas printed in the lecture. Lecture 30 page 17 gives a direct clearing-time formula only when electrical power is zero during the fault; that condition is not imposed here.

6 Test Case

6.1 Network and Chosen Data

MATPOWER case9 supplies the topology, reactances, charging, loads and generation schedules. Branch resistances are set to zero following Lecture 30. All values below use a 100 MVA base.

Branch	R1	X1	Total B1
1-4	0	.0576	0
4-5	0	.092	.158
5-6	0	.17	.358
3-6	0	.0586	0
6-7	0	.1008	.209
7-8	0	.072	.149
8-2	0	.0625	0
8-9	0	.161	.306
9-4	0	.085	.176

At 100% loading, buses 5, 7 and 9 consume 90+j30, 100+j35 and 125+j50 MVA. G2 and G3 supply 163 and 85 MW. G1 is the slack. Generator terminal voltages are 1 pu. Frequency is chosen as 50 Hz; transmission voltage base is 345 kV. Initial mechanical powers are 0.67, 1.63 and 0.85 pu.

Chosen parameter	G1	G2	G3
H on system base (s)	23.64	6.4	3.01
Xd' (pu)	.0608	.1198	.1813
Damping	0	0	0

Supplemental study inputs, not supplied in case9 or the slides: generator X2 = Xd'; line Z2 = Z1, Z0 = 3Z1 and B0 = B1/3; generator-side delta and transmission-side grounded-wye transformers; zero-sequence-open loads with equal positive/negative sequence impedances. These choices make the unbalanced-fault model reproducible.

6.2 Duration Scan

Scan 0-3 s in 0.05 s increments. Use RK4 with 0.002 s steps and exact switching instants; observe 20 s after clearing. A run fails if any unwrapped pairwise rotor-angle separation reaches 360 degrees. If a passing/failing interval is found, bisect it to at most 0.001 s. This finite-horizon indicator does not prove asymptotic stability.

6.3 Program Output

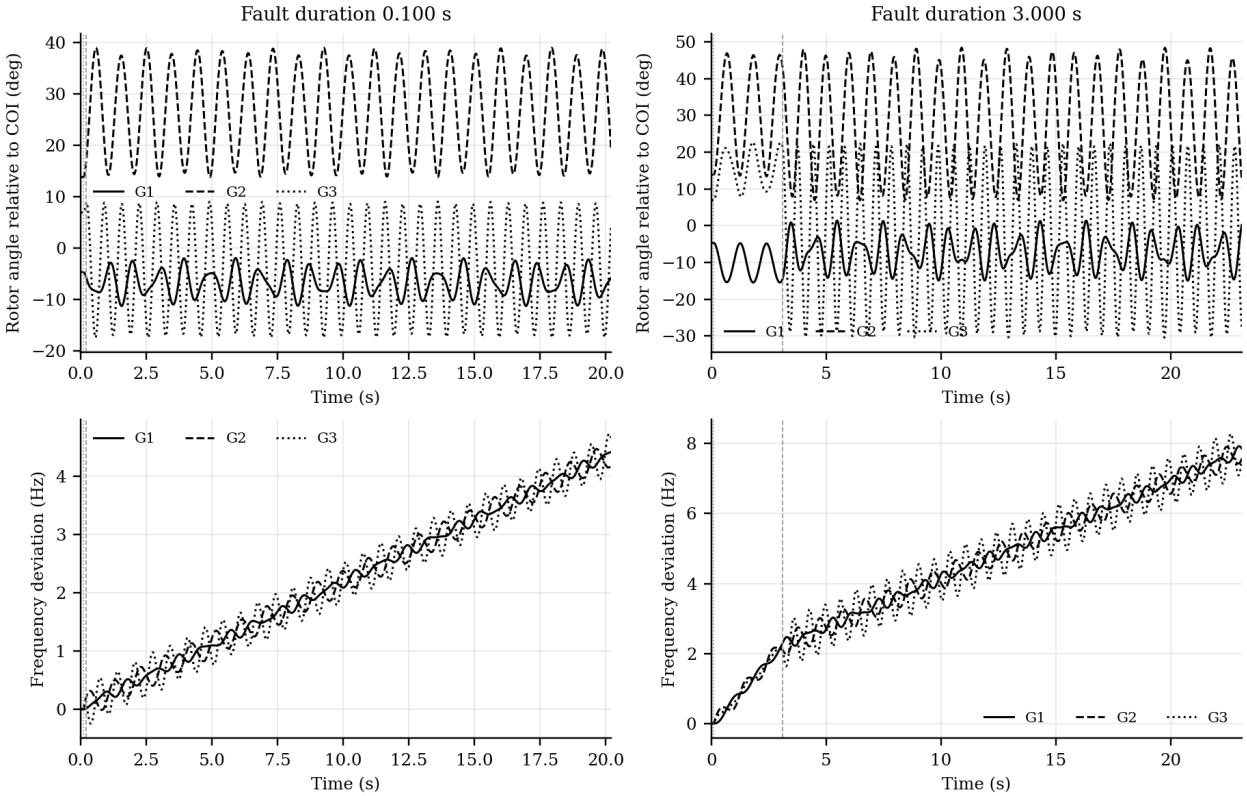


Figure 1. Fault durations 0.100 s and 3.000 s at normal loading. Both trials pass the stated relative-angle criterion. Curves use the supplied JavaScript reference solver; they are not screenshots of a MATLAB run.

Quantity	**Result**
Fault resistance	7 ohm
Loading factor	1.0
Duration range / spacing	0-3 s / 0.05 s
Number of sampled durations	61
Passing durations	61
Finite CCT	Not identified in the sampled range
Observation after clearing	20 s
Numerical angle threshold	360 degrees

All sampled durations pass; this does not imply infinite CCT and does not prove that every unsampled duration passes.

6.4 Verification

The complete duration scan was repeated with the integration step reduced from 2 ms to 1 ms, and with a 40 s post-clear observation horizon. No boundary was found in either repeated scan. An independent Newton load flow and adaptive DOP853 integration also passed trials at 0, 0.1, 1 and 3 s. The reconstructed fault condition satisfies Va = Rf Ia.

Check	Result
Independent power-flow mismatch	Below 10^-10 pu
Independent SLG boundary residual	Below 10^-10 pu
Half-step complete scan	No boundary in 0-3 s
Longer-observation complete scan	No boundary in 0-3 s
Native MATLAB execution	Not available during preparation

Run the supplied .m file locally to produce MATLAB-native figures and results. Python and JavaScript agreement validates the implemented model independently, but is not a substitute for an actual MATLAB run.

7 Conclusion

A nine-bus model was constructed with the undamped classical-machine assumptions in Lecture 30 and the sequence-network treatment of the SLG fault. For Rf = 7 ohm, normal loading and the specified line 7-8 clearing action, all 61 sampled fault durations from 0 to 3 s pass the chosen relative-angle test. A finite critical clearing time was not identified within this search.

The answer depends on the chosen network, inertia, sequence data, fault location and clearing action. Governors, exciters, electromagnetic transients, negative-sequence heating and torque ripple are omitted. Voltage and reactive-power limits are not enforced. A passing rotor-angle test does not establish frequency security. Common frequency drift grows without governors; the long-horizon curves are checks of this fixed-frequency model and have limited physical accuracy.

Sources

EE 315 Lecture 30 (28 October): pages 3-4, swing equation and common-base inertia; page 7, model assumptions; page 17, restricted clearing-time formula. Lecture 27 (16 October): single-line-to-ground sequence networks. Lecture 31 (30 October): single-machine equal-area examples. Assignment 10: nine-bus transient study and Rf rule.

Network data: MATPOWER case9. Dynamic and sequence inputs are declared above; they are not attributed to the slides or to case9.